



Orazio Pontorno

AI RESEARCHER

opontorno.github.io

Italian
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CONTACT

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RESEARCH AREAS

- Computer Vision
- Image Processing
- AIGC Detection
- Generative Models
- Applied Mathematics
- Multimedia
- Media Forensics
- Graph Learning
- Anomaly Detection

TOOLS & TECH

- Languages
- Python
 - R
 - SQL
 - MATLAB
- ML & DL
- PyTorch
 - scikit-learn
 - NumPy
 - pandas
 - MLflow
- Data & Cloud
- PySpark
 - Databricks
 - Amazon SageMaker
 - MySQL
- Workflow
- Git
 - LaTeX
 - Docker
 - Tableau

LANGUAGES

Italian Native
English B2

INTERESTS

Chess · Fitness · Travel

ABOUT

Final-year PhD researcher in Artificial Intelligence with a solid theoretical foundation in Applied Mathematics, holding a Master's in Data Science and a Bachelor's in Mathematics. Specializes in theoretical and methodological advancements within Computer Vision, Image Processing, Synthetic Media Forensics, and AIGC Detection. Experienced in conducting cutting-edge scientific research across international academic institutions and corporate R&D environments.

EXPERIENCE

- Jan 2026 - Jun 2026
Visiting Researcher
SUNY Polytechnic Institute · Utica, NY, USA · On-site · Full-time
Deepfake analysis and detection in medical imaging.
Medical Image Analysis · AIGC Detection · Computer Vision · Generative Models · Image Processing
- Mar 2025 - Dec 2025
AI Researcher
Life360 · San Mateo, CA, USA · Remote · Full-time
ML models for personalized ad targeting and causal inference in recommendation systems.
Causal Inference · Similarity Models · Databricks · Amazon SageMaker AI · PySpark
- Apr 2024 - Feb 2025
R&D Data Scientist
Fantix Inc. · Milan, Italy · Hybrid · Contract
Data enrichment models based on graph and hypergraph representations.
Graph Neural Networks · Hypergraph Models · Amazon SageMaker AI · PyTorch · Deep Learning
- Apr 2023 - Jun 2023
Student Research Assistant
iCTLab s.r.l. · Catania, Italy · Hybrid · Internship
Deep learning models for synthetic media forensics and deepfake detection.
Multimedia · Deepfake Detection · Generative Models · Image Processing · Deep Learning
- Feb 2023 - Aug 2023
Research Assistant
Vicosistems s.r.l. · Catania, Italy · Hybrid · Scholarship
GAN-based anomaly detection on time-series data for predictive maintenance.
Anomaly Detection · Time-series · Predictive Maintenance · GAN-networks · Deep Learning

EDUCATION

- Nov 2023 - present
PhD in Artificial Intelligence - XXXIX Cycle
Università Campus Bio-Medico of Roma · University of Catania
- Oct 2021 - Sep 2023
Master's Degree in Data Science
University of Catania
Grade: 110/110 summa cum laude
Thesis: An AI Forensics approach for the recognition of synthetic image-generating architecture based on diffusion models
- Sep 2022 - Nov 2022
Postgraduate Course - AI: Deep Learning, Vision & Language
Università degli Studi di Modena e Reggio Emilia
- Sep 2018 - Dec 2021
Bachelor's Degree in Mathematics
University of Catania
Grade: 105/110
Thesis: Teoria dei Codici: Codici Lineari

SELECTED PUBLICATIONS

- 2026 - European Conference on Computer Vision (ECCV)
μFlow: Leveraging Average Images for Improving Generalisation of Deepfake Faces Detectors
Authors: O. Pontorno*, M. Litrico*, L. Guanera, V. Giuffrida, S. Battiato
Project page: opontorno.github.io/MuFlow
doi: 10.1007/978-3-032-37422-6_20

■ 2026 - 34th ACM International Conference on Multimedia

HexMIL: Hierarchical Attention MIL for Ante-Hoc Explainable Detection of AI-Manipulated CT Volumes

Authors: **O. Pontorno**, L. Guanera, Z. Akhtar, S. Battiato

🔗 Project page: opontorno.github.io/hexmil

📄 doi: 10.1145/3767308.3836040

■ 2025 - *Multimedia Tools and Application*

DeepFeatureX-SN: Generalization of deepfake detection via contrastive learning

Authors: **O. Pontorno**, L. Guanera, S. Battiato

📄 doi: 10.1007/s11042-025-21060-1

■ 2024 - *IEEE International Conference on Image Processing (ICIP)*

On the exploitation of DCT-traces in the generative-AI domain

Authors: **O. Pontorno**, L. Guanera, S. Battiato

📄 doi: 10.1109/ICIP51287.2024.10648013

Catania, 5th October 2026


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